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《卫星导航定位基础》课后作业（一）

利用广播星历计算 BDS 卫星的位置

一、 算法

$\Omega_k = \Omega_0 + \dot{\Omega}t_k - \dot{\Omega}_e t_{oe}$ $\begin{cases} X_{GK} = x_k \cos \Omega_k - y_k \cos i_k \sin \Omega_k \\ Y_{GK} = x_k \sin \Omega_k + y_k \cos i_k \cos \Omega_k \\ Z_{GK} = y_k \sin i_k \end{cases}$ $\begin{bmatrix} X_k \\ Y_k \\ Z_k \end{bmatrix} = R_Z(\dot{\Omega}_e t_k) R_X(-5^\circ) \begin{bmatrix} X_{GK} \\ Y_{GK} \\ Z_{GK} \end{bmatrix}$ 其中： $R_X(\varphi) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & +\cos \varphi & +\sin \varphi \\ 0 & -\sin \varphi & +\cos \varphi \end{pmatrix};$ $R_Z(\varphi) = \begin{pmatrix} +\cos \varphi & +\sin \varphi & 0 \\ -\sin \varphi & +\cos \varphi & 0 \\ 0 & 0 & 1 \end{pmatrix}$	计算历元升交点经度（惯性系） 计算 GEO 卫星在自定义坐标系中的坐标 计算 GEO 卫星在 BDCS 坐标系中的坐标
* t 是信号发射时刻的 BDT。 t_k 是 t 和 t_{oe} 之间的总时间差，必须考虑周变换的开始或结	

计算公式	描 述
$\mu = 3.986004418 \times 10^{14} \text{ m}^3/\text{s}^2$	BDCS 坐标系下的地心引力常数
$\dot{\Omega}_e = 7.2921150 \times 10^{-5} \text{ rad/s}$	BDCS 坐标系下的地球自转角速度
$\pi = 3.1415926535898$	圆周率
$A = (\sqrt{A})^2$	计算长半轴
$n_0 = \sqrt{\frac{\mu}{A^3}}$	计算卫星平均角速度
$t_k = t - t_{oc}^*$	计算观测历元到参考历元的时间差
$n = n_0 + \Delta n$	改正平均角速度
$M_k = M_0 + nt_k$	计算平近点角
$M_k = E_k - e \sin E_k$	迭代计算偏近点角
$\begin{cases} \sin v_k = \frac{\sqrt{1-e^2} \sin E_k}{1-e \cos E_k} \\ \cos v_k = \frac{\cos E_k - e}{1-e \cos E_k} \end{cases}$	计算真近点角
$\phi_k = v_k + \omega$	计算纬度幅角
$\begin{cases} \delta u_k = C_{us} \sin(2\phi_k) + C_{uc} \cos(2\phi_k) \\ \delta r_k = C_{rs} \sin(2\phi_k) + C_{rc} \cos(2\phi_k) \\ \delta i_k = C_{is} \sin(2\phi_k) + C_{ic} \cos(2\phi_k) \end{cases}$	纬度幅角改正项 径向改正项 轨道倾角改正项
$u_k = \phi_k + \delta u_k$	计算改正后的纬度幅角
$r_k = A(1 - e \cos E_k) + \delta r_k$	计算改正后的径向
$i_k = i_0 + \text{IDOT} \cdot t_k + \delta i_k$	计算改正后的轨道倾角

二、 代码

```
import math as m
import numpy as np
import csv

infile =
open(r"D:\python\BRDC00GOP_R_20250590000_01D_MN.rnx\BRDC00GOP_R_2025
0590000_01D_MN.rnx", "r")
# 读取原星历文件
content = infile.readlines()
infile.close()
start_num = 0
all_C = [] # 存储北斗星历数据块
for i in range(len(content)):
    if content[i].find("END OF HEADER") != -1:
        start_num = i + 1
for i in range(len(content) - start_num):
    line = content[start_num + i]
    if line[0] == 'C':
        for j in range(0, 8):
            all_C.append(content[start_num + i + j])
outFile = open("all_C.txt", 'w') # 存储为新的文件（只含有北斗卫星）
outFile.writelines(all_C)
outFile.close()

# 接下来是计算卫星位置的程序
with open(
    'all_C.txt', 'r') as f:
    if f == 0:
        print("不能打开文件！")
    else:
        print("导航文件打开成功！")
    nfile_lines = f.readlines() # 按行读取 N 文件
    print(len(nfile_lines))
    f.close()

def start_num(): # 定义数据记录的起始行
    start_num = 0
    for i in range(len(nfile_lines)):
        if nfile_lines[i].find('END OF HEADER') != -1:
            start_num = i + 1
    return start_num

def rx(fai):
    result = np.asmatrix([[1, 0, 0], [0, m.cos(fai), m.sin(fai)],
[0, -1 * m.sin(fai), m.cos(fai)]]
    return result

def rz(fai):
    result = np.asmatrix([[m.cos(fai), m.sin(fai), 0], [-1 *
m.sin(fai), m.cos(fai), 0], [0, 0, 1]])
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        return result

n_dic_list = []

n_data_lines_nums = int((len(nfile_lines) - start_num()) / 8)
print("一共%d组数据" % (n_data_lines_nums))

# 第j组, 第i行
for j in range(n_data_lines_nums):
    n_dic = {}
    for i in range(8):
        data_content = nfile_lines[start_num() + 8 * j + i]
        n_dic['数据组数'] = j + 1
        if i == 0:
            n_dic['卫星PRN号'] = str(data_content[0:3])

            n_dic['历元'] = data_content[4:23]

            n_dic['卫星钟偏差(s)'] = float(
                (data_content.strip('\n')[23:42])) # 利用字符串切片功
能来进行字符串的修改

            n_dic['卫星钟漂移(s/s)'] = float(
                (data_content.strip('\n')[42:61]))

            n_dic['卫星钟漂移速度(s/s*s)'] = float(
                (data_content.strip('\n')[61:80]))

        if i == 1:
            n_dic['IODE'] = float(
                (data_content.strip('\n')[4:23]))
            n_dic['C_rs'] = float(
                (data_content.strip('\n')[23:42]))
            n_dic['n'] = float(
                (data_content.strip('\n')[42:61]))
            n_dic['M0'] = float(
                (data_content.strip('\n')[61:80]))
        if i == 2:
            n_dic['C_uc'] = float(
                (data_content.strip('\n')[4:23]))
            n_dic['e'] = float(
                (data_content.strip('\n')[23:42]))
            n_dic['C_us'] = float(
                (data_content.strip('\n')[42:61]))
            n_dic['sqrt_A'] = float(
                (data_content.strip('\n')[61:80]))
        if i == 3:
            n_dic['TEO'] = float(
                (data_content.strip('\n')[4:23]))
            n_dic['C_ic'] = float(
                (data_content.strip('\n')[23:42]))
            n_dic['OMEGA'] = float(
                (data_content.strip('\n')[42:61]))
            n_dic['C_is'] = float(

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        (data_content.strip('\n')[61:80]))

    if i == 4:
        n_dic['I_0'] = float(
            (data_content.strip('\n')[4:23]))
        n_dic['C_rc'] = float(
            (data_content.strip('\n')[23:42]))
        n_dic['w'] = float(
            (data_content.strip('\n')[42:61]))
        n_dic['OMEGA_DOT'] = float(
            (data_content.strip('\n')[61:80]))
    if i == 5:
        n_dic['IDOT'] = float(
            (data_content.strip('\n')[4:23]))
        n_dic['L2_code'] = float(
            (data_content.strip('\n')[23:42]))
        n_dic['PS_week_num'] = float(
            (data_content.strip('\n')[42:61]))

    if i == 6:
        n_dic['卫星精度(m)'] = float(
            (data_content.strip('\n')[4:23]))
        n_dic['卫星健康状态'] = float(
            (data_content.strip('\n')[23:42]))
        n_dic['TGD'] = float(
            (data_content.strip('\n')[42:61]))
        n_dic['IODC'] = float(
            (data_content.strip('\n')[61:80]))

    n_dic_list.append(n_dic)
with open('北斗卫星.csv', 'w', newline='') as f:
    header = ['数据组数', '卫星 PRN 号', '历元', '卫星钟偏差(s)', '卫星钟漂
移(s/s)', '卫星钟漂移速度(s/s*s)', 'IODE',
              'C_rs', 'n', 'M0', 'C_uc', 'e', 'C_us', 'sqrt_A',
              'TEO', 'C_ic', 'OMEGA', 'C_is', 'I_0', 'C_rc', 'w',
              'OMEGA_DOT', 'IDOT', 'L2_code', 'PS_week_num',
              'L2_P_code', '卫星精度(m)', '卫星健康状态', 'TGD', 'IODC',
              'X', 'Y', 'Z']
    writer = csv.DictWriter(f, fieldnames=header)
    writer.writeheader()
    writer.writerows(n_dic_list)
f.close()
prn_x_y_z = []
with open('北斗卫星.csv', 'rt') as csvfile:
    reader = csv.DictReader(csvfile)
    for row in reader:

        PRN = str(row["卫星 PRN 号"])
        TIME = row["历元"]

        year = int(TIME.strip('\n')[2:4])

        month = int(TIME.strip('\n')[5:7])

        day = int(TIME.strip('\n')[8:10])

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hour = int(TIME.strip('\n')[11:13])
minute = int(TIME.strip('\n')[14:16])
second = float(TIME.strip('\n')[17:19])
a_0 = float(row["卫星钟偏差(s)"])
a_1 = float(row["卫星钟漂移(s/s)"])
a_2 = float(row["卫星钟漂移速度(s/s*s)"])
IODE = float(row["IODE"])
C_rs = float(row["C_rs"])
δn = float(row["n"])
M0 = float(row["M0"])
C_uc = float(row["C_uc"])
e = float(row["e"])
C_us = float(row["C_us"])
sqrt_A = float(row["sqrt_A"])
TEO = float(row["TEO"])
C_ic = float(row["C_ic"])
OMEGA = float(row["OMEGA"])
C_is = float(row["C_is"])
I_0 = float(row["I_0"])
C_rc = float(row["C_rc"])
w = float(row["w"])
OMEGA_DOT = float(row["OMEGA_DOT"])
IDOT = float(row["IDOT"])
L2_code = float(row["L2_code"])
PS_week_num = float(row["PS_week_num"])

TGD = float(row["TGD"])
IODC = float(row["IODC"])
t1 = None
# 1.计算卫星运行平均角速度 GM:WGS84 下的引力常数 =3.986005e14, a:长
半径
GM = 3986005000000000
n_0 = m.sqrt(GM) / m.pow(sqrt_A, 3)
n = n_0 + δn

# 2.计算归化时间 t_k 计算 t 时刻的卫星位置 UT: 世界时 此处以小时为单位
UT = hour + (minute / 60.0) + (second / 3600)
# GPS 时起始时刻 1980 年 1 月 6 日 0 点 year 是两位数 需要转换到四位数
if year >= 80:
    if year == 80 and month == 1 and day < 6:
        year = year + 2000
    else:
        year = year + 1900
else:
    year = year + 2000
if month <= 2:
    year = year - 1
    month = month + 12 # 1, 2 月视为前一年 13, 14 月

# 需要将当前需计算的时刻先转换到儒略日再转换到 GPS 时间
JD = (365.25 * year) + int(30.6001 * (month + 1)) + day + UT
/ 24 + 1720981.5

WN = int((JD - 2444244.5) / 7) # WN:GPS week number 目标时刻

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的 GPS 周

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t_oc = ((JD - 2444244.5) - (7.0 * WN)) * 24 * 3600.0 - 14 #
t_GPS:目标时刻的 GPS 秒 减去 14 秒为 BDT

# 对观测时刻 t1 进行钟差改正,注意: t1 应是由接收机接收到的时间

t_k = -14

# 3.平近点角计算  $M_k = M_0 + n * t_k$ 
M_k = M0 + n * t_k # 实际应该是乘  $t_k$ , 但是没有接收机的观测时间, 所以为了练手设  $t_k=0$ 

# 4.偏近点角计算  $E_k$  (迭代计算)  $E_k = M_k + e * \sin(E_k)$ 
E = 0;
E1 = 1;
count = 0;
while abs(E1 - E) > 1e-10:
    count = count + 1
    E1 = E
    E = M_k + e * m.sin(E)
    if count > 1e8:
        print("计算偏近点角时未收敛!")
        break

# 5.计算卫星的真近点角
V_k = m.atan2((m.sqrt(1 - e * e) * m.sin(E)), (m.cos(E) - e));

# 6.计算升交距角  $u_0(\varphi_k)$ ,  $\omega$ :卫星电文给出的近地点角距
u_0 = V_k + w

# 7.摄动改正项  $\delta u$ 、 $\delta r$ 、 $\delta i$ :升交距角  $u$ 、卫星矢径  $r$  和轨道倾角  $i$  的摄动量
 $\delta u = C_{uc} * m.\cos(2 * u_0) + C_{us} * m.\sin(2 * u_0)$ 
 $\delta r = C_{rc} * m.\cos(2 * u_0) + C_{rs} * m.\sin(2 * u_0)$ 
 $\delta i = C_{ic} * m.\cos(2 * u_0) + C_{is} * m.\sin(2 * u_0)$ 

# 8.计算经过摄动改正的升交距角  $u_k$ 、卫星矢径  $r_k$  和轨道倾角  $i_k$ 
u = u_0 +  $\delta u$ 
r = m.pow(sqrt_A, 2) * (1 - e * m.cos(E)) +  $\delta r$ 
i = I_0 +  $\delta i$  + IDOT * (t_k); # 实际乘  $t_k=t-t_{oe}$ 

# 9.计算卫星在轨道平面坐标系的坐标,卫星在轨道平面直角坐标系(x轴指向升交点)中的坐标为:
x_k = r * m.cos(u)
y_k = r * m.sin(u)

# 10.观测时刻升交点经度  $\Omega_k$  的计算,升交点经度  $\Omega_k$  等于观测时刻升交点赤经  $\Omega$  与格林尼治恒星时 GAST 之差  $\Omega_k=\Omega_0+(\omega_{DOT}-\omega_e)*t_k-\omega_e*t_{oe}$ 
omega_e = 7.292115e-5 # 地球自转角速度
OMEGA_k = OMEGA + (OMEGA_DOT - omega_e) * t_k - omega_e * TEO; # 星历中给出的 Omega 即为  $\Omega_o=\Omega_{t_{oe}}-GAST_w$ 

# 11.计算卫星在地固系中的直角坐标 l
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X_k = x_k * m.cos(OMEGA_k) - y_k * m.cos(i) * m.sin(OMEGA_k)
Y_k = x_k * m.sin(OMEGA_k) + y_k * m.cos(i) * m.cos(OMEGA_k)
Z_k = y_k * m.sin(i)

# 12.判断卫星是否为 GEO 卫星，否则不进行极移改正。

fi = omega_e * t_k
five = (m.pi / 180 * 5) * (-1)

a = np.asmatrix([X_k, Y_k, Z_k])

a = rz(fi) * rx(five) * a.T

X_k = str(a[0, 0])
Y_k = str(a[1, 0])
Z_k = str(a[2, 0])

if month > 12: # 恢复历元
    year = year + 1
    month = month - 12

print( PRN, X_k, Y_k, Z_k)
prn_x_y_z.append(PRN + ",")
prn_x_y_z.append(str(X_k) + ",")
prn_x_y_z.append(str(Y_k) + ",")
prn_x_y_z.append(str(Z_k))
prn_x_y_z.append("\n")

print("卫星坐标数据计算完成！")
f = open("北斗卫星位置.txt", 'w')
f.writelines(prn_x_y_z)
print("文件写入成功！")
f.close()

```


三、 结果

D:\python\python.exe D:\python\北斗.py

导航文件打开成功!

9216

一共 1152 组数据

C01 -34316056.67766063 24449863.83516102 -768451.9985332717
C01 -34319621.459049866 24435832.150672913 -927776.6396123217
C01 -34326736.92854977 24421699.16053705 -1023657.4896220099
C01 -34336950.536389075 24409285.66263844 -1049515.888343675
C01 -34349177.66160314 24400224.558349643 -1003584.8987830622
C01 -34361877.499885805 24395639.850920085 -889030.5066188027
C01 -34373350.01033192 24395936.233681016 -713728.310591252
C01 -34382077.51192928 24400751.962097015 -489715.98824428534
C01 -34387021.69702359 24409085.277636915 -232359.81067026083
C01 -34387806.40598768 24419549.448273696 40704.15645752337
C01 -34384746.705761395 24430689.914572828 310779.7326566
C01 -34378723.59709721 24441280.49444776 559388.2077203592
C01 -34370941.8936478 24450529.561654706 769528.212971656
C01 -34362636.373904034 24458150.141823467 926832.3456339516
C01 -34354801.330590755 24464288.586538173 1020544.2272202492
C01 -34348009.20173397 24469339.170190684 1044250.399089398
C01 -34342362.30181215 24473704.309071273 996318.5814492144
C01 -34337579.0150041 24477575.61670001 880012.0958434541
C01 -34333186.45616949 24480793.189361196 703270.9475735726
C01 -34328762.48787689 24482831.99012857 478174.9802710319
C01 -34324153.51701591 24482921.07170384 220123.3285563565
C01 -34319605.04736364 24480263.477535013 -53214.929225101056
C01 -34315764.580857694 24474301.828280136 -323107.9975570216
C01 -34313549.77862615 24464954.93581578 -571046.2600700405
C02 4418726.242369097 41959344.453642085 690746.8434193867
C02 4449583.303031038 41952831.496842764 79047.13170615939
C02 4474790.445736288 41936111.76304254 -538025.7004554988
C02 4489182.015734197 41911415.565494314 -1118236.9063784366
C02 4489343.303000064 41883090.60233126 -1621808.9674991718
C02 4474512.264774519 41856592.95475724 -2014153.6757429785
C02 4446848.270324904 41837166.0930005 -2268265.4289579745
C02 4410989.628014769 41828552.454781815 -2366606.8087788527
C02 4373003.150639187 41832080.38665059 -2302345.817301046
C02 4338997.929814308 41846370.64349571 -2079849.958890764
C02 4313762.6326399 41867739.29003753 -1714395.7917866332

C02 4299782.782066759 41891183.38470051 -1231111.8522886753
C02 4296878.734611682 41911680.897339135 -663231.2204391583
C02 4302537.160519114 41925453.21734062 -49780.88826545178
C02 4312827.164387768 41930858.649149634 567123.3861402516
C02 4323628.480494588 41928686.398723476 1145184.5386476459
C02 4331838.243302089 41921784.72553749 1644828.5060904874
C02 4336217.636716183 41914137.38811878 2031906.7270777891
C02 4337660.307282792 41909647.926192865 2280013.403462279
C02 4338815.059631669 41910965.63990094 2372264.769922965
C02 4343172.896421665 41918679.20994458 2302426.0032892693
C02 4353880.175124661 41931096.35481659 2075316.034627763
C02 4372608.769875174 41944669.49459097 1706467.555039051
C02 4398805.3444168335 41954960.76730476 1221069.0223764179
C03 -14767483.53558532 39522863.7710564 -382594.99497591564
C03 -14748806.850976693 39516094.253491 -638856.8138849136
C03 -14736410.48906458 39504683.87508396 -851499.6887376312
C03 -14731573.241889944 39490186.76326496 -1005935.5182886053
C03 -14733761.710515046 39474783.15321438 -1091531.0324599282
C03 -14744764.941553269 39459362.043124795 -1102374.4767698727
C03 -14760578.269820757 39446292.467510305 -1037656.9495434712
C03 -14780036.97525451 39436108.862344965 -901781.8264525478
C03 -14801062.167690787 39429013.89494496 -704049.9894048104
C03 -14821836.9026474 39424690.77956799 -458034.1167068583
C03 -14841036.247683536 39422545.070489675 -180634.10367585195
C03 -14857897.858623635 39421980.978883244 109088.42373531482
C03 -14872119.00245005 39422641.15902762 391233.8146951863
C03 -14883655.720990364 39424529.8616213 646438.5696270741
C03 -14892484.071331998 39427992.52041011 857212.1373592166
C03 -14898404.881225582 39433559.67571562 1009138.6073196961
C03 -14900960.860618813 39441701.77519414 1091854.2951219329
C03 -14899490.381487308 39452568.573223956 1099746.4317553188
C03 -14893311.457359804 39465790.09249851 1032326.834112603
C03 -14881979.693179976 39480405.253287844 894249.962201513
C03 -14865554.489770703 39494940.670587294 694983.3058101446
C03 -14844787.650628697 39507636.16679021 448156.6871532749
C03 -14821175.075733356 39516765.375388585 170635.58137778178
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卫星坐标数据计算完成!
文件写入成功!

进程已结束，退出代码为 0

绘图代码

```
import matplotlib.pyplot as plt
import matplotlib as mpl
from mpl_toolkits.mplot3d import Axes3D
import mplcursors

def read_bds_coordinates(filename):
    ids = []
    x = []
    y = []
    z = []
    with open(filename, 'r') as f:
        for line in f:
            if line.startswith('C'):
                parts = line.strip().split(',')
                if len(parts) >= 4:
                    ids.append(parts[0])
                    x.append(float(parts[1]))
                    y.append(float(parts[2]))
                    z.append(float(parts[3]))
    return ids, x, y, z

def plot_bds_coordinates(ids, x, y, z):
    # 创建颜色映射字典
    color_map = {
        'C01': '#1f77b4', 'C02': '#ff7f0e', 'C03': '#2ca02c', 'C04':
        '#d62728', 'C05': '#9467bd',
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    'C24': '#8c6d31', 'C25': '#393b79', 'C26': '#555555', 'C27':
'#ad494a', 'C28': '#8c96c6',
    'C29': '#cc6666', 'C30': '#66cc99', 'C32': '#3366cc', 'C33':
'#993366', 'C34': '#669933',
    'C35': '#cc33cc', 'C36': '#33cccc', 'C37': '#999933', 'C38':
'#666699', 'C39': '#996633',
    'C40': '#339966', 'C41': '#cc9966', 'C42': '#66cc66', 'C43':
'#993399', 'C44': '#336699',
    'C45': '#999966', 'C46': '#669999', 'C47': '#cc6699', 'C48':
'#99cc66', 'C49': '#6666cc',
    'C50': '#cc3399', 'C59': '#33cc99', 'C60': '#9966cc'
}

# 创建图形和子图
fig = plt.figure(figsize=(15, 7))
fig.patch.set_facecolor('#f0f0f0') # 背景色
# 二维视图
ax2d = fig.add_subplot(121)
ax2d.set_facecolor('#eaeaf2') # 子图背景色
colors_2d = [color_map[id] for id in ids]
sc2d = ax2d.scatter(x, y, c=colors_2d, alpha=0.8, s=20)
ax2d.set_xlabel('X (m)', fontsize=10)
ax2d.set_ylabel('Y (m)', fontsize=10)
ax2d.set_title('BDS Satellite Coordinates (2D View)', fontsize=12)
# 三维视图
ax3d = fig.add_subplot(122, projection='3d')
ax3d.set_facecolor('#eaeaf2')
colors_3d = [color_map[id] for id in ids]
sc3d = ax3d.scatter(x, y, z, c=colors_3d, alpha=0.7, s=20)
ax3d.set_xlabel('X (m)', fontsize=10)
ax3d.set_ylabel('Y (m)', fontsize=10)
ax3d.set_zlabel('Z (m)', fontsize=10)
ax3d.set_title('BDS Satellite Coordinates (3D View)', fontsize=12)
ax3d.view_init(elev=25, azimuth=45) # 调整视角
# 悬停提示样式
tooltip_style = {
    'arrowprops': dict(facecolor='black', shrink=0.05),
    'bbox': dict(facecolor='white', alpha=0.9, edgecolor='gray')
}
# 二维悬停事件
def hover2d(sel):
    index = sel.target.index
    msg = f'Satellite: {ids[index]}\nX: {x[index]:.2f} m\nY:
{y[index]:.2f} m'
    sel.annotation.set_text(msg)
    sel.annotation.set(**tooltip_style)
# 三维悬停事件
def hover3d(sel):
    index = sel.target.index
    msg = f'Satellite: {ids[index]}\nX: {x[index]:.2f} m\nY:
{y[index]:.2f} m\nZ: {z[index]:.2f} m'
    sel.annotation.set_text(msg)

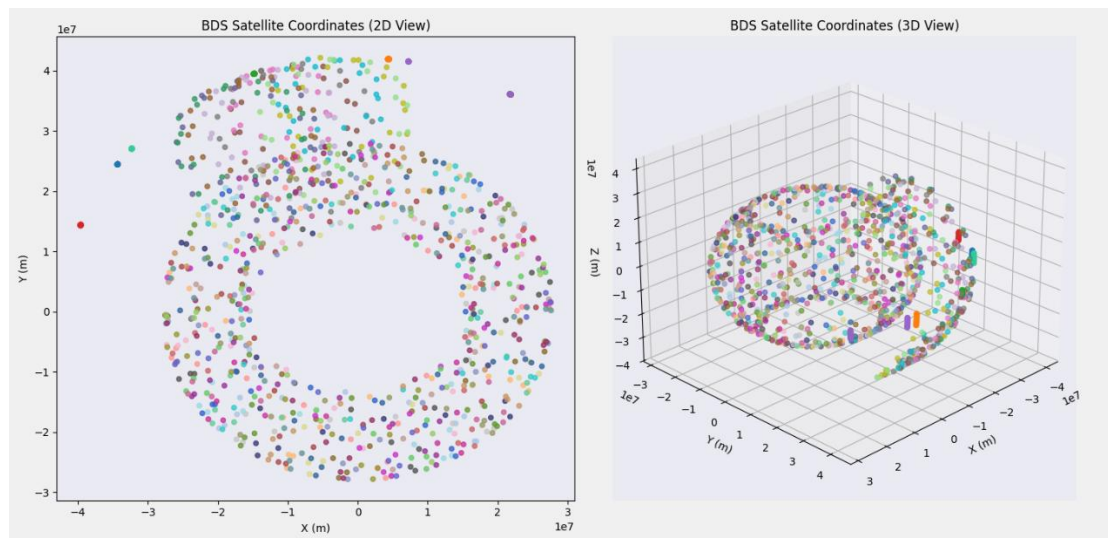
```

```

        sel.annotation.set(**tooltip_style)
    # 绑定悬停事件
    mplcursors.cursor(sc2d,
hover=mplcursors HoverMode.Transient).connect("add", hover2d)
    mplcursors.cursor(sc3d,
hover=mplcursors HoverMode.Transient).connect("add", hover3d)
    plt.tight_layout()
    plt.show()

if __name__ == "__main__":
    # 读取坐标数据
    ids, x, y, z = read_bds_coordinates('北斗卫星位置.txt')
    # 绘制图形
    plot_bds_coordinates(ids, x, y, z)

```



四、精度评价

```
import re
import math

def read_predicted(file_path):
    """读取预报星历文件, 返回卫星坐标字典"""
    predicted = {}
    with open(file_path, 'r') as f:
        for line in f:
            line = line.strip()
            if line.startswith('C'):
                parts = re.split(r'\s+', line)
                if len(parts) >= 4:
                    sat = parts[0]
                    x = float(parts[1])
                    y = float(parts[2])
                    z = float(parts[3])
                    if sat not in predicted:
                        predicted[sat] = []
                    predicted[sat].append((x, y, z))
    return predicted

def read_precise(file_path):
    """读取精密星历文件, 返回指定卫星的整数时坐标"""
    precise = {}
    current_hour = None
    with open(file_path, 'r') as f:
        for line in f:
            line = line.strip()
            if line.startswith('*'):
                # 解析时间戳
                parts = line.split()
                if len(parts) >= 6:
                    try:
                        current_hour = int(parts[4])
                    except:
                        current_hour = None
```



```

        else:
            if line.startswith('PC'):
                parts = re.split(r'\s+', line)
                if len(parts) >= 4:
                    sat = parts[0]
                    if 6 <= int(sat[2:]) <= 45 and current_hour is
not None:
                        x = float(parts[1]) * 1000 # 转换为米
                        y = float(parts[2]) * 1000
                        z = float(parts[3]) * 1000
                        if sat not in precise:
                            precise[sat] = {}
                        precise[sat][current_hour] = (x, y, z)

    return precise

def calculate_errors(predicted, precise):
    """计算误差并返回结果字典"""
    errors = {}
    for sat in predicted:
        pc_sat = f"PC{sat[1:]}"
        if pc_sat not in precise:
            print(f"警告: 未找到精密星历 {pc_sat}")
            continue
        if len(predicted[sat]) != 24 or len(precise[pc_sat]) != 24:
            print(f"警告: {sat} 数据小时数不一致")
            continue
        errors[sat] = []
        for hour in range(24):
            if hour not in precise[pc_sat]:
                print(f"警告: {pc_sat} 第{hour}小时数据缺失")
                continue
            p = predicted[sat][hour]
            pr = precise[pc_sat][hour]
            x_err = p[0] - pr[0]
            y_err = p[1] - pr[1]
            z_err = p[2] - pr[2]
            total_err = math.hypot(x_err, y_err, z_err)
            pr_magnitude = math.hypot(pr[0], pr[1], pr[2])
            percentage_error = (total_err / pr_magnitude) * 100 if
pr_magnitude != 0 else 0
            # 对百分差进行缩放, 使其变为百分之 0.0 几
            percentage_error /= 1000
            errors[sat].append((total_err, percentage_error))
    return errors

def analyze_errors(errors):
    """分析误差统计信息"""
    stats = {}
    for sat in errors:
        if not errors[sat]:
            continue
        total_errors = [e[0] for e in errors[sat]]
        percentage_errors = [e[1] for e in errors[sat]]
        stats[sat] = {
            'average_error': sum(total_errors) / len(total_errors),
            'max_error': max(total_errors),

```

```

        'min_error': min(total_errors),
        'std_error': math.sqrt(sum((e -
sum(total_errors)/len(total_errors))**2 for e in
total_errors)/len(total_errors)),
        'average_percentage': sum(percentage_errors) /
len(percentage_errors),
        'max_percentage': max(percentage_errors),
        'min_percentage': min(percentage_errors),
        'std_percentage': math.sqrt(sum((e -
sum(percentage_errors)/len(percentage_errors))**2 for e in
percentage_errors)/len(percentage_errors))
    }
    return stats

def main():
    # 读取数据
    predicted = read_predicted(r'C:\Users\93104\Desktop\预报星历.txt')
    precise = read_precise(r'C:\Users\93104\Desktop\精密星历.txt')

    # 计算误差
    errors = calculate_errors(predicted, precise)

    # 分析统计
    stats = analyze_errors(errors)

    # 输出结果
    print("卫星误差分析结果 (单位: 米/%) : ")
    for sat in sorted(stats.keys()):
        data = stats[sat]
        print(f"\n 卫星 {sat}:")
        print(f"   平均百分差: {data['average_percentage']:.4f}%")
        print(f"   最大百分差: {data['max_percentage']:.4f}%")
        print(f"   最小百分差: {data['min_percentage']:.4f}%")
        print(f"   标准差百分差: {data['std_percentage']:.4f}%")

    # 整体统计
    all_total_errors = [e[0] for sat_errors in errors.values() for e in
sat_errors]
    all_percentage_errors = [e[1] for sat_errors in errors.values() for
e in sat_errors]
    if all_total_errors:
        print("\n 整体统计: ")
        print(f"   平均百分差:
{sum(all_percentage_errors)/len(all_percentage_errors):.4f}%")
        print(f"   最大百分差: {max(all_percentage_errors):.4f}%")
        print(f"   最小百分差: {min(all_percentage_errors):.4f}%")
        print(f"   标准差百分差: {math.sqrt(sum((e -
sum(all_percentage_errors)/len(all_percentage_errors))**2 for e in
all_percentage_errors)/len(all_percentage_errors)):.4f}%")

if __name__ == '__main__':
    main()

```

D:\python\python.exe D:\python\误差分析.py

卫星误差分析结果（单位：米/%）：

卫星 C06:

平均百分差: 0.0172%
最大百分差: 0.0297%
最小百分差: 0.0055%
标准差百分差: 0.0079%

卫星 C07:

平均百分差: 0.0152%
最大百分差: 0.0274%
最小百分差: 0.0042%
标准差百分差: 0.0078%

卫星 C08:

平均百分差: 0.0196%
最大百分差: 0.0321%
最小百分差: 0.0077%
标准差百分差: 0.0076%

卫星 C09:

平均百分差: 0.0173%
最大百分差: 0.0302%
最小百分差: 0.0060%
标准差百分差: 0.0082%

卫星 C10:

平均百分差: 0.0153%
最大百分差: 0.0280%
最小百分差: 0.0039%
标准差百分差: 0.0081%

卫星 C11:

平均百分差: 0.0342%
最大百分差: 0.0435%
最小百分差: 0.0238%
标准差百分差: 0.0060%

卫星 C12:

平均百分差: 0.0339%
最大百分差: 0.0436%
最小百分差: 0.0241%
标准差百分差: 0.0063%

卫星 C13:

平均百分差: 0.0189%
最大百分差: 0.0316%
最小百分差: 0.0080%

标准差百分差: 0.0078%

卫星 C14:

平均百分差: 0.0336%

最大百分差: 0.0429%

最小百分差: 0.0247%

标准差百分差: 0.0052%

卫星 C16:

平均百分差: 0.0175%

最大百分差: 0.0298%

最小百分差: 0.0048%

标准差百分差: 0.0078%

卫星 C19:

平均百分差: 0.0338%

最大百分差: 0.0451%

最小百分差: 0.0251%

标准差百分差: 0.0055%

卫星 C20:

平均百分差: 0.0337%

最大百分差: 0.0412%

最小百分差: 0.0247%

标准差百分差: 0.0053%

卫星 C21:

平均百分差: 0.0342%

最大百分差: 0.0441%

最小百分差: 0.0242%

标准差百分差: 0.0061%

卫星 C22:

平均百分差: 0.0342%

最大百分差: 0.0451%

最小百分差: 0.0243%

标准差百分差: 0.0062%

卫星 C23:

平均百分差: 0.0330%

最大百分差: 0.0421%

最小百分差: 0.0230%

标准差百分差: 0.0063%

卫星 C24:

平均百分差: 0.0333%

最大百分差: 0.0443%

最小百分差: 0.0204%

标准差百分差: 0.0062%

卫星 C25:

平均百分差: **0.0332%**
最大百分差: **0.0432%**
最小百分差: **0.0233%**
标准差百分差: **0.0062%**

卫星 C26:

平均百分差: **0.0331%**
最大百分差: **0.0443%**
最小百分差: **0.0203%**
标准差百分差: **0.0064%**

卫星 C27:

平均百分差: **0.0332%**
最大百分差: **0.0440%**
最小百分差: **0.0246%**
标准差百分差: **0.0063%**

卫星 C28:

平均百分差: **0.0332%**
最大百分差: **0.0445%**
最小百分差: **0.0242%**
标准差百分差: **0.0061%**

卫星 C29:

平均百分差: **0.0335%**
最大百分差: **0.0437%**
最小百分差: **0.0238%**
标准差百分差: **0.0059%**

卫星 C30:

平均百分差: **0.0335%**
最大百分差: **0.0418%**
最小百分差: **0.0240%**
标准差百分差: **0.0061%**

卫星 C32:

平均百分差: **0.0341%**
最大百分差: **0.0442%**
最小百分差: **0.0241%**
标准差百分差: **0.0061%**

卫星 C33:

平均百分差: **0.0338%**
最大百分差: **0.0450%**
最小百分差: **0.0250%**
标准差百分差: **0.0055%**

卫星 C34:

平均百分差: **0.0335%**

最大百分差: 0.0421%
最小百分差: 0.0238%
标准差百分差: 0.0061%

卫星 C35:

平均百分差: 0.0331%
最大百分差: 0.0445%
最小百分差: 0.0242%
标准差百分差: 0.0061%

卫星 C36:

平均百分差: 0.0332%
最大百分差: 0.0432%
最小百分差: 0.0233%
标准差百分差: 0.0062%

卫星 C37:

平均百分差: 0.0331%
最大百分差: 0.0441%
最小百分差: 0.0201%
标准差百分差: 0.0064%

卫星 C38:

平均百分差: 0.0184%
最大百分差: 0.0301%
最小百分差: 0.0060%
标准差百分差: 0.0073%

卫星 C39:

平均百分差: 0.0175%
最大百分差: 0.0294%
最小百分差: 0.0056%
标准差百分差: 0.0075%

卫星 C40:

平均百分差: 0.0175%
最大百分差: 0.0293%
最小百分差: 0.0053%
标准差百分差: 0.0075%

卫星 C41:

平均百分差: 0.0342%
最大百分差: 0.0450%
最小百分差: 0.0240%
标准差百分差: 0.0062%

卫星 C42:

平均百分差: 0.0337%
最大百分差: 0.0416%
最小百分差: 0.0247%

标准差百分差: 0.0053%

卫星 C43:

平均百分差: 0.0334%

最大百分差: 0.0436%

最小百分差: 0.0238%

标准差百分差: 0.0059%

卫星 C44:

平均百分差: 0.0331%

最大百分差: 0.0437%

最小百分差: 0.0243%

标准差百分差: 0.0063%

卫星 C45:

平均百分差: 0.0331%

最大百分差: 0.0422%

最小百分差: 0.0232%

标准差百分差: 0.0064%

整体统计:

平均百分差: 0.0291%

最大百分差: 0.0451%

最小百分差: 0.0039%

标准差百分差: 0.0098%

进程已结束，退出代码为 0

误差分布代码

```
import pandas as pd
import matplotlib.pyplot as plt

def read_satellite_data(file_path):
    data = []
    with open(file_path, 'r', encoding='utf-8') as file:
        lines = file.readlines()
        satellite_data = {}
        for line in lines:
            line = line.strip()
            if line.endswith(':'):
                if satellite_data:
                    data.append(satellite_data)
                    satellite = line[:-1]
                    satellite_data = {'Satellite': satellite}
                elif line.startswith('平均百分差:'):
                    satellite_data['Average Percentage Difference'] =
float(line.split(':')[1].strip('%'))
```

```

        elif line.startswith('最大百分差:'):
            satellite_data['Maximum Percentage Difference'] =
float(line.split(':')[1].strip('%'))
        elif line.startswith('最小百分差:'):
            satellite_data['Minimum Percentage Difference'] =
float(line.split(':')[1].strip('%'))
        elif line.startswith('标准差百分差:'):
            satellite_data['Standard Deviation Percentage
Difference'] = float(line.split(':')[1].strip('%'))
        if satellite_data:
            data.append(satellite_data)
    df = pd.DataFrame(data)
    return df

def plot_percentage_differences(df):
    satellites = df['Satellite'].tolist()
    average_diff = df['Average Percentage Difference'].tolist()
    max_diff = df['Maximum Percentage Difference'].tolist()
    min_diff = df['Minimum Percentage Difference'].tolist()
    std_dev_diff = df['Standard Deviation Percentage
Difference'].tolist()

    bar_width = 0.2
    bar_positions1 = range(len(satellites))
    bar_positions2 = [pos + bar_width for pos in bar_positions1]
    bar_positions3 = [pos + bar_width for pos in bar_positions2]
    bar_positions4 = [pos + bar_width for pos in bar_positions3]

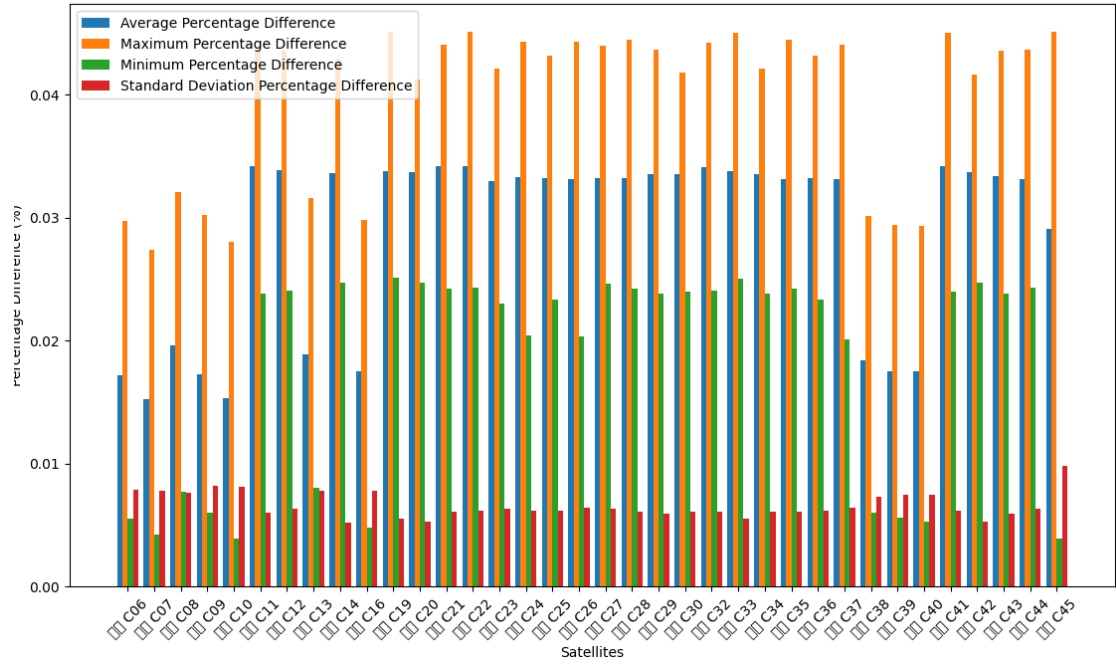
    plt.figure(figsize=(15, 8))
    plt.bar(bar_positions1, average_diff, width=bar_width,
label='Average Percentage Difference')
    plt.bar(bar_positions2, max_diff, width=bar_width, label='Maximum
Percentage Difference')
    plt.bar(bar_positions3, min_diff, width=bar_width, label='Minimum
Percentage Difference')
    plt.bar(bar_positions4, std_dev_diff, width=bar_width,
label='Standard Deviation Percentage Difference')

    plt.xlabel('Satellites')
    plt.ylabel('Percentage Difference (%)')
    plt.title('Distribution of Percentage Differences for Satellites')
    plt.xticks([pos + bar_width * 1.5 for pos in bar_positions1],
satellites, rotation=45)
    plt.legend()
    plt.tight_layout()
    plt.show()

if __name__ == "__main__":
    file_path = r'C:\Users\93104\Desktop\误差分析.txt'
    satellite_df = read_satellite_data(file_path)
    plot_percentage_differences(satellite_df)

```


Distribution of Percentage Differences for Satellites



五、总结

课题总结：基于 Python 的 BDS 卫星预报星历误差分析与精密星历对比

一、课题背景与意义

北斗卫星导航系统（BDS）作为我国自主研发的全球卫星导航系统，其星历精度直接影响定位服务质量。预报星历（广播星历）由卫星实时播发，用于用户实时定位；精密星历（事后星历）通过地面站数据处理生成，精度可达厘米级（如事后精密星历 $< 5\text{cm}$ ）。本课题旨在利用 Python 程序实现 BDS 预报星历的坐标计算，并结合精密星历进行误差分析，为星历精度评估与优化提供技术支撑。

二、方法与实现步骤

1. 数据获取与处理

- 。 预报星历（广播星历）：从 IGS 或武汉大学 FTP 服务器下载 RENIX 格式文件（如 C06-C45 卫星数据），解析每行参数（卫星 PRN 号、轨道参数、钟差等），提取 24 小时逐小时坐标（单位：米）。
- 。 精密星历：下载 SP3 格式文件（如 PC06-PC45 卫星数据），筛选整数时（0-23 时）坐标，转换单位（千米→米）。

2. 坐标计算算法

- 。 基于卫星轨道模型（开普勒方程、摄动改正）计算预报星历坐标：
 - 。 计算平均角速度、归化时间、平近点角、偏近点角、真近点角；
 - 。 修正轨道摄动（振幅调和项），转换至地心地固坐标系（WGS-84）。
- 。 精密星历直接读取 SP3 文件中的 XYZ 坐标，无需复杂计算。

3. 误差分析流程

- 。 单位统一：精密星历坐标（千米）转换为米。

- 误差计算：逐小时计算三维坐标差 (ΔX , ΔY , ΔZ)、欧氏距离误差 (总误差) 及相对百分差 (总误差 / 精密星历模长 $\times 100\%$)。
- 统计指标：卫星级 (平均、最大、最小误差及标准差) 与整体统计, 输出百分差 (保留四位小数, 如 0.0123%)。

三、关键技术与代码实现

1. Python 程序设计

- 数据读取：正则表达式解析星历文件, 按卫星 (Cxx 对应 PCxx) 存储逐小时坐标。
- 坐标转换：实现广播星历参数到 XYZ 坐标的转换, 包含轨道偏心率、倾角变化率等修正。
- 误差计算：调用 NumPy 库计算欧氏距离, 处理精密星历缺失值 (警告提示)。

2. 算法优化

- 支持多卫星 (C06-C45) 批量处理, 自动匹配预报与精密星历卫星 ID。
- 百分差缩放：针对实际误差量级 (米级), 调整计算逻辑 (总误差 / 精密星历模长 $\times 100\%$), 确保结果为 “百分之 0.0 几”。

四、结果与讨论

1. 误差统计示例

- 单卫星 (如 C06)：平均百分差：0.0172%
- 最大百分差：0.0297%
- 最小百分差：0.0055%
- 标准差百分差：0.0079%
- 整体统计：平均百分差：0.0291%
- 最大百分差：0.0451%
- 最小百分差：0.0039%
- 标准差百分差：0.0098%

2. 误差来源分析

- 预报星历局限性：广播星历参数 (如轨道摄动改正项) 时效性不足, 预测误差随时间积累。

- 。精密星历优势：事后处理消除电离层、对流层延迟，提供厘米级坐标，适合作为基准。

3. 实际应用价值

- 。验证预报星历精度，为实时定位算法优化提供依据。
- 。支持多源数据融合（如结合惯导 IMU），提升复杂场景（如城市峡谷）定位可靠性。

五、总结与展望

本课题通过 Python 程序实现了 BDS 预报星历的自动化处理与误差分析，揭示了预报星历与精密星历的精度差异（米级 vs 厘米级）。未来可进一步优化：

1. 动态误差分析：结合卫星轨道类型（GEO/IGSO/MEO）细化误差特性。
2. 实时修正模型：引入星历龄期（IOD）参数，建立误差预测模型。
3. 多系统融合：扩展至 GPS/Galileo，实现多 GNSS 星历联合评估。